Finally, we note that MATLAB's random numbers are only seemingly unpredictable. In fact, MATLAB maintains a seed value that determines the subsequent "random" numbers that will be returned. This seed is controlled by the `rng` function; `s=rng` saves the current seed and `rng(s)` restores a previously saved seed. Initializing the random number generator with the same seed always generates the same sequence:

Example 1.26

```
>> s=rng;
>> 50+randi(50,1,12)
ans =
   89   76   80   80   72   92   58   56   77   78   59   58
>> rng(s);
>> 50+randi(50,1,12)
ans =
   89   76   80   80   72   92   58   56   77   78   59   58
```

When you run a simulation that uses `rand`, it normally doesn't matter how the `rng` seed is initialized. However, it can be instructive to use the same repeatable sequence of `rand` values when you are debugging your simulation.

Quiz 1.7

The number of characters in a tweet is equally likely to be any integer between 1 and 140. Simulate an experiment that generates 1000 tweets and counts the number of "long" tweets that have over 120 characters. Repeat this experiment 5 times.

Problems

Difficulty: ● Easy ■ Moderate ◆ Difficult ◆◆ Experts Only

1.1.1● Continuing Quiz 1.1, write Gerlanda's entire menu in words (supply prices if you wish).

1.1.2● For Gerlanda's pizza in Quiz 1.1, answer these questions:

(a) Are N and M mutually exclusive?

(b) Are N, T, and M collectively exhaustive?

(c) Are T and O mutually exclusive? State this condition in words.

(d) Does Gerlanda's make Tuscan pizzas with mushrooms and onions?

(e) Does Gerlanda's make Neapolitan pizzas that have neither mushrooms nor onions?

1.1.3● Ricardo's offers customers two kinds of pizza crust, Roman (R) and Neapolitan (N). All pizzas have cheese but not all pizzas have tomato sauce. Roman pizzas can have tomato sauce or they can be white (W); Neapolitan pizzas always have tomato sauce. It is possible to order a Roman pizza with mushrooms (M) added. A Neapolitan pizza can contain mushrooms or onions (O) or both, in addition to the tomato sauce and cheese. Draw a Venn diagram that shows the relationship among the ingredients N, M, O, T, and W in the menu of Ricardo's pizzeria.

1.2.1● A hypothetical wi-fi transmission can take place at any of three speeds depending on the condition of the radio channel between a laptop and an access point. The speeds are high (h) at 54 Mb/s, medium (m) at 11 Mb/s, and low (l) at 1 Mb/s. A user of the wi-fi connection can transmit a short signal corresponding to a mouse click (c), or a long signal corresponding to a tweet (t). Consider the experiment of monitoring wi-fi signals and observing the transmission speed and the length. An observation is a two-letter word, for example, a high-speed, mouse-click transmission is hm.

(a) What is the sample space of the experiment?

(b) Let A_1 be the event "medium speed connection." What are the outcomes in A_1?

(c) Let A_2 be the event "mouse click." What are the outcomes in A_2?

(d) Let A_3 be the event "high speed connection or low speed connection." What are the outcomes in A_3?

(e) Are A_1, A_2, and A_3 mutually exclusive?

(f) Are A_1, A_2, and A_3 collectively exhaustive?

1.2.2 An integrated circuit factory has three machines X, Y, and Z. Test one integrated circuit produced by each machine. Either a circuit is acceptable (a) or it fails (f). An observation is a sequence of three test results corresponding to the circuits from machines X, Y, and Z, respectively. For example, aaf is the observation that the circuits from X and Y pass the test and the circuit from Z fails the test.

(a) What are the elements of the sample space of this experiment?

(b) What are the elements of the sets

$$Z_F = \{\text{circuit from } Z \text{ fails}\},$$
$$X_A = \{\text{circuit from } X \text{ is acceptable}\}.$$

(c) Are Z_F and X_A mutually exclusive?

(d) Are Z_F and X_A collectively exhaustive?

(e) What are the elements of the sets

$$C = \{\text{more than one circuit acceptable}\},$$
$$D = \{\text{at least two circuits fail}\}.$$

(f) Are C and D mutually exclusive?

(g) Are C and D collectively exhaustive?

1.2.3 Shuffle a deck of cards and turn over the first card. What is the sample space of this experiment? How many outcomes are in the event that the first card is a heart?

1.2.4 Find out the birthday (month and day but not year) of a randomly chosen person. What is the sample space of the experiment? How many outcomes are in the event that the person is born in July?

1.2.5 The sample space of an experiment consists of all undergraduates at a university. Give four examples of partitions.

1.2.6 The sample space of an experiment consists of the measured resistances of two resistors. Give four examples of partitions.

1.3.1 Find $P[B]$ in each case:

(a) Events A and B are a partition and $P[A] = 3\,P[B]$.

(b) For events A and B, $P[A \cup B] = P[A]$ and $P[A \cap B] = 0$.

(c) For events A and B, $P[A \cup B] = P[A] - P[B]$.

1.3.2 You roll two fair six-sided dice; one die is red, the other is white. Let R_i be the event that the red die rolls i. Let W_j be the event that the white die rolls j.

(a) What is $P[R_3 W_2]$?

(b) What is the $P[S_5]$ that the sum of the two rolls is 5?

1.3.3 You roll two fair six-sided dice. Find the probability $P[D_3]$ that the absolute value of the difference of the dice is 3.

1.3.4 Indicate whether each statement is true or false.

(a) If $P[A] = 2\,P[A^c]$, then $P[A] = 1/2$.

(b) For all A and B, $P[AB] \leq P[A]\,P[B]$.

(c) If $P[A] < P[B]$, then $P[AB] < P[B]$.

(d) If $P[A \cap B] = P[A]$, then $P[A] \geq P[B]$.

1.3.5● Computer programs are classified by the length of the source code and by the execution time. Programs with more than 150 lines in the source code are big (B). Programs with ≤ 150 lines are little (L). Fast programs (F) run in less than 0.1 seconds. Slow programs (W) require at least 0.1 seconds. Monitor a program executed by a computer. Observe the length of the source code and the run time. The probability model for this experiment contains the following information: $P[LF] = 0.5$, $P[BF] = 0.2$, and $P[BW] = 0.2$. What is the sample space of the experiment? Calculate the following probabilities: $P[W]$, $P[B]$, and $P[W \cup B]$.

1.3.6● There are two types of cellular phones, handheld phones (H) that you carry and mobile phones (M) that are mounted in vehicles. Phone calls can be classified by the traveling speed of the user as fast (F) or slow (W). Monitor a cellular phone call and observe the type of telephone and the speed of the user. The probability model for this experiment has the following information: $P[F] = 0.5$, $P[HF] = 0.2$, $P[MW] = 0.1$. What is the sample space of the experiment? Find the following probabilities $P[W]$, $P[MF]$, and $P[H]$.

1.3.7● Shuffle a deck of cards and turn over the first card. What is the probability that the first card is a heart?

1.3.8● You have a six-sided die that you roll once and observe the number of dots facing upwards. What is the sample space? What is the probability of each sample outcome? What is the probability of E, the event that the roll is even?

1.3.9● A student's score on a 10-point quiz is equally likely to be any integer between 0 and 10. What is the probability of an A, which requires the student to get a score of 9 or more? What is the probability the student gets an F by getting less than 4?

1.3.10■ Use Theorem 1.4 to prove the following facts:

(a) $P[A \cup B] \geq P[A]$

(b) $P[A \cup B] \geq P[B]$

(c) $P[A \cap B] \leq P[A]$

(d) $P[A \cap B] \leq P[B]$

1.3.11■ Use Theorem 1.4 to prove by induction the *union bound*: For any collection of events $A_1, \ldots, A_n$,

$$P[A_1 \cup A_2 \cup \cdots \cup A_n] \leq \sum_{i=1}^{n} P[A_i].$$

1.3.12◆ Using *only* the three axioms of probability, prove $P[\varnothing] = 0$.

1.3.13◆ Using the three axioms of probability and the fact that $P[\varnothing] = 0$, prove Theorem 1.3. Hint: Define $A_i = B_i$ for $i = 1, \ldots, m$ and $A_i = \varnothing$ for $i > m$.

1.3.14◆◆ For each fact stated in Theorem 1.4, determine which of the three axioms of probability are needed to prove the fact.

1.4.1● Mobile telephones perform *handoffs* as they move from cell to cell. During a call, a telephone either performs zero handoffs (H_0), one handoff (H_1), or more than one handoff (H_2). In addition, each call is either long (L), if it lasts more than three minutes, or brief (B). The following table describes the probabilities of the possible types of calls.

	H_0	H_1	H_2
L	0.1	0.1	0.2
B	0.4	0.1	0.1

(a) What is the probability that a brief call will have no handoffs?

(b) What is the probability that a call with one handoff will be long?

(c) What is the probability that a long call will have one or more handoffs?

1.4.2● You have a six-sided die that you roll once. Let R_i denote the event that the roll is i. Let G_j denote the event that the roll is greater than j. Let E denote the event that the roll of the die is even-numbered.

(a) What is $P[R_3|G_1]$, the conditional probability that 3 is rolled given that the roll is greater than 1?

(b) What is the conditional probability that 6 is rolled given that the roll is greater than 3?

(c) What is $P[G_3|E]$, the conditional probability that the roll is greater than 3 given that the roll is even?

(d) Given that the roll is greater than 3, what is the conditional probability that the roll is even?

1.4.3● You have a shuffled deck of three cards: 2, 3, and 4. You draw one card. Let C_i denote the event that card i is picked. Let E denote the event that the card chosen is a even-numbered card.

(a) What is $P[C_2|E]$, the probability that the 2 is picked given that an even-numbered card is chosen?

(b) What is the conditional probability that an even-numbered card is picked given that the 2 is picked?

1.4.4■ Phonesmart is having a sale on Bananas. If you buy one Banana at full price, you get a second at half price. When couples come in to buy a pair of phones, sales of Apricots and Bananas are equally likely. Moreover, given that the first phone sold is a Banana, the second phone is twice as likely to be a Banana rather than an Apricot. What is the probability that a couple buys a pair of Bananas?

1.4.5■ The basic rules of genetics were discovered in mid-1800s by Mendel, who found that each characteristic of a pea plant, such as whether the seeds were green or yellow, is determined by two genes, one from each parent. In his pea plants, Mendel found that yellow seeds were a dominant trait over green seeds. A yy pea with two yellow genes has yellow seeds; a gg pea with two recessive genes has green seeds; a hybrid gy or yg pea has yellow seeds. In one of Mendel's experiments, he started with a parental generation in which half the pea plants were yy and half the plants were gg. The two groups were crossbred so that each pea plant in the first generation was gy. In the second generation, each pea plant was equally likely to inherit a y or a g gene from each first-generation parent. What is the probability $P[Y]$ that a randomly chosen pea plant in the second generation has yellow seeds?

1.4.6■ From Problem 1.4.5, what is the conditional probability of yy, that a pea plant has two dominant genes given the event Y that it has yellow seeds?

1.4.7■ You have a shuffled deck of three cards: 2, 3, and 4, and you deal out the three cards. Let E_i denote the event that ith card dealt is even numbered.

(a) What is $P[E_2|E_1]$, the probability the second card is even given that the first card is even?

(b) What is the conditional probability that the first two cards are even given that the third card is even?

(c) Let O_i represent the event that the ith card dealt is odd numbered. What is $P[E_2|O_1]$, the conditional probability that the second card is even given that the first card is odd?

(d) What is the conditional probability that the second card is odd given that the first card is odd?

1.4.8♦ Deer ticks can carry both Lyme disease and human granulocytic ehrlichiosis (HGE). In a study of ticks in the Midwest, it was found that 16% carried Lyme disease, 10% had HGE, and that 10% of the ticks that had either Lyme disease or HGE carried both diseases.

(a) What is the probability $P[LH]$ that a tick carries both Lyme disease (L) and HGE (H)?

(b) What is the conditional probability that a tick has HGE given that it has Lyme disease?

1.5.1● Given the model of handoffs and call lengths in Problem 1.4.1,

(a) What is the probability $P[H_0]$ that a phone makes no handoffs?

(b) What is the probability a call is brief?

(c) What is the probability a call is long or there are at least two handoffs?

1.5.2● For the telephone usage model of Example 1.18, let B_m denote the event that a call is billed for m minutes. To generate a phone bill, observe the duration of the call in integer minutes (rounding up). Charge for M minutes $M = 1, 2, 3, \ldots$ if the exact duration T is $M - 1 < t \le M$. A more complete probability model shows that for $m = 1, 2, \ldots$ the probability of each event B_m is

$$P[B_m] = \alpha(1-\alpha)^{m-1}$$

where $\alpha = 1 - (0.57)^{1/3} = 0.171$.

(a) Classify a call as long, L, if the call lasts more than three minutes. What is $P[L]$?

(b) What is the probability that a call will be billed for nine minutes or less?

1.5.3◆ Suppose a cellular telephone is equally likely to make zero handoffs (H_0), one handoff (H_1), or more than one handoff (H_2). Also, a caller is either on foot (F) with probability 5/12 or in a vehicle (V).

(a) Given the preceding information, find three ways to fill in the following probability table:

	H_0	H_1	H_2
F			
V			

(b) Suppose we also learn that 1/4 of all callers are on foot making calls with no handoffs and that 1/6 of all callers are vehicle users making calls with a single handoff. Given these additional facts, find all possible ways to fill in the table of probabilities.

1.6.1● Is it possible for A and B to be independent events yet satisfy $A = B$?

1.6.2● Events A and B are equiprobable, mutually exclusive, and independent. What is $P[A]$?

1.6.3● At a Phonesmart store, each phone sold is twice as likely to be an Apricot as a Banana. Also each phone sale is independent of any other phone sale. If you monitor the sale of two phones, what is the probability that the two phones sold are the same?

1.6.4■ Use a Venn diagram in which the event areas are proportional to their probabilities to illustrate two events A and B that are independent.

1.6.5■ In an experiment, A and B are mutually exclusive events with probabilities $P[A] = 1/4$ and $P[B] = 1/8$.

(a) Find $P[A \cap B]$, $P[A \cup B]$, $P[A \cap B^c]$, and $P[A \cup B^c]$.

(b) Are A and B independent?

1.6.6■ In an experiment, C and D are independent events with probabilities $P[C] = 5/8$ and $P[D] = 3/8$.

(a) Determine the probabilities $P[C \cap D]$, $P[C \cap D^c]$, and $P[C^c \cap D^c]$.

(b) Are C^c and D^c independent?

1.6.7■ In an experiment, A and B are mutually exclusive events with probabilities $P[A \cup B] = 5/8$ and $P[A] = 3/8$.

(a) Find $P[B]$, $P[A \cap B^c]$, and $P[A \cup B^c]$.

(b) Are A and B independent?

1.6.8■ In an experiment, C, and D are independent events with probabilities $P[C \cap D] = 1/3$, and $P[C] = 1/2$.

(a) Find $P[D]$, $P[C \cap D^c]$, and $P[C^c \cap D^c]$.

(b) Find $P[C \cup D]$ and $P[C \cup D^c]$.

(c) Are C and D^c independent?

1.6.9■ In an experiment with equiprobable outcomes, the sample space is $S = \{1, 2, 3, 4\}$ and $P[s] = 1/4$ for all $s \in S$. Find three events in S that are pairwise independent but are not independent. (Note:

Pairwise independent events meet the first three conditions of Definition 1.7).

1.6.10■ (Continuation of Problem 1.4.5) One of Mendel's most significant results was the conclusion that genes determining different characteristics are transmitted independently. In pea plants, Mendel found that round peas (r) are a dominant trait over wrinkled peas (w). Mendel crossbred a group of (rr, yy) peas with a group of (ww, gg) peas. In this notation, rr denotes a pea with two "round" genes and ww denotes a pea with two "wrinkled" genes. The first generation were either (rw, yg), (rw, gy), (wr, yg), or (wr, gy) plants with both hybrid shape and hybrid color. Breeding among the first generation yielded second-generation plants in which genes for each characteristic were equally likely to be either dominant or recessive. What is the probability $P[Y]$ that a second-generation pea plant has yellow seeds? What is the probability $P[R]$ that a second-generation plant has round peas? Are R and Y independent events? How many visibly different kinds of pea plants would Mendel observe in the second generation? What are the probabilities of each of these kinds?

1.6.11◆ For independent events A and B, prove that

(a) A and B^c are independent.

(b) A^c and B are independent.

(c) A^c and B^c are independent.

1.6.12◆ Use a Venn diagram in which the event areas are proportional to their probabilities to illustrate three events A, B, and C that are independent.

1.6.13◆ Use a Venn diagram in which event areas are in proportion to their probabilities to illustrate events A, B, and C that are pairwise independent but not independent.

1.7.1● Following Quiz 1.3, use MATLAB, but not the `randi` function, to generate a vector $\mathbf{T}$ of 200 independent test scores such that all scores between 51 and 100 are equally likely.